

Global Smooth Path Planning for Mobile Robots based on Continuous Bezier Curve

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Abstract—In this paper, a new strategy combining the MDPSO (multimodal delayed particle swarm optimization) and continuous Bezier curve is developed for the global smooth path planning of mobile robots. Firstly, the preliminaries on Bezier curve and the MDPSO are briefly introduced. Secondly, the environment modeling is presented and the smooth path planning problem is mathematically formulated. Then, the new strategy is applied to devise an optimum smooth global path with continuous curvature. In the simulation experiments, the developed strategy has been verified to be able to successfully complete the challenging task of planning global smooth path, and the generated smooth path could outperform the ones in previous studies. Finally, the paper is concluded with conclusions and future works.

I. INTRODUCTION

Path planning is one of the critical issues of mobile robots whose objective is to devise a feasible and optimal route from the start to the target position. Such an issue can be taken as an optimization problem upon certain given performance indices and constraints [1]. To date, a large number of heuristic algorithms (e.g. genetic algorithm and particle swarm optimization) have been proposed to deal with this optimization problem, see e.g. [2]–[4] and the references therein. However, only some simple criteria and essential constraints (e.g. the minimum path length and collision-free route) have been considered in the implementation of the path planning, where the generated paths are usually composed of several polygonal lines. To follow these paths, the mobile robots would have to switch frequently between different motion states (e.g. stop, rotation and restart), which will lead to unnecessary waste of time and energy. Specially, the jerk motion of the state switching could not be permissible when the smooth movement is required to ensure some service qualities [5]. Generally, the path smoothness has been proven to be another significant criterion with close relation to other optimization criteria [6], and the issue of smooth path planning of mobile robots is increasingly attracting particular research attentions in recent years.

In the past decades, several parametric curves have been developed and combined with intelligent optimization algo-

rithms to tackle the issues of smooth path planning. For example, GA (genetic algorithm) combining with Bezier curve is applied to plan a smooth global path, where the smoothness is directly linked to the criteria of the path optimization [7]. A stochastic PSO (particle swarm optimization) algorithm has been proposed to devise an optimal smooth cubic spline path, where the coefficients of a cubic polynomial spline compose the particles of the particle swarm [8]. In [9], a parallel PSO algorithm has been put forward to plan the global path for mobile robots, where the generated linear path is subsequently smoothed by a cubic B-spline path smoother. A new strategy combining PSO with Bezier curve is proposed to plan smooth global path for mobile robots in [10], where the Bezier curve segments are connected to make a smooth path. Nevertheless, there are at least the following drawbacks with the above approaches: 1) the generated smooth paths are usually produced by connecting several smooth path segments, so that the continuities (e.g. continuous curvature and curvature derivative) are difficult to be satisfied at the joint points of the path; 2) the constrained conditions (e.g. bounded curvature and minimum curvature derivative) are seldom concerned with the optimization criteria of the smooth path planning; 3) the optimization of smooth path is very likely to be trapped into local optima, which could be a suboptimum but infeasible path.

To handle the issues mentioned above, a new strategy combining an improved PSO algorithm and continuous Bezier curve is developed to plan global smooth path for mobile robots in this paper. The main contributions of this paper are threefold. 1) *a rigorous mathematical formulation of the smooth path planning optimization problem is formulated*; 2) *a continuous Bezier curve is employed to plan a smooth global path, so that the continuity of curvature on the path is satisfied*; 3) *the minimum curvature is linked to the optimization indices and an improved PSO algorithm is successful applied to tackle with the frequently appearing local trapping phenomenon of this issue*.

The rest of this paper is organized as follows. In section II, the preliminaries on Bezier curve is briefly introduced, and the MDPSO algorithm is provided to deal with the planning issues of smooth path. In section III, the environment modeling and problem statement are presented, and then the MDPSO combining with Bezier curve are employed to plan smooth

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global path for mobile robots. Finally, the conclusions and future works of this paper are presented in the last section.

II. PRELIMINARIES

A. Bezier curve

The Bezier curve is a parametric curve that is frequently used in computer graphics and other related fields [11]. Given a series of control points, the corresponding Bezier curve (denoted as $\mathbf{p}(t)$) is defined as

$$\mathbf{p}(t) = \sum_{i=0}^n b_i^n(t) \mathbf{p}_i, \quad 0 \leq t \leq 1, \quad (1)$$

where t indicates the normalized time variable; $\mathbf{p}_i = (x_i, y_i)^T$ denotes the coordinate vector of the i th control point, i.e. X and Y coordinate components; $b_i^n(t)$ stands for the Bernstein polynomial that is expressed as

$$b_i^n(t) = \binom{n}{i} t^i (1-t)^{n-i}, \quad i = 0, 1, \dots, n. \quad (2)$$

Accordingly, the first derivatives of the Bezier curve can be calculated as

$$\dot{\mathbf{p}}(t) = \frac{d\mathbf{p}(t)}{dt} = n \sum_{i=0}^{n-1} b_i^{n-1}(t) (\mathbf{p}_{i+1} - \mathbf{p}_i), \quad (3)$$

and the higher derivatives of the Bezier curve can be calculated by using equation (3) iteratively. Thus the Bezier curve's curvature can be calculated as

$$\kappa(t) = \frac{\dot{\mathbf{p}}_x(t) \ddot{\mathbf{p}}_y(t) - \dot{\mathbf{p}}_y(t) \ddot{\mathbf{p}}_x(t)}{(\dot{\mathbf{p}}_x^2(t) + \dot{\mathbf{p}}_y^2(t))^{3/2}}, \quad (4)$$

where $\dot{\mathbf{p}}_x(t)$, $\dot{\mathbf{p}}_y(t)$, $\ddot{\mathbf{p}}_x(t)$ and $\ddot{\mathbf{p}}_y(t)$ are respectively the corresponding coordinate components of the first and second derivatives of the Bezier curve $\mathbf{p}(t)$.

B. Improved PSO Algorithm

Particle swarm optimization (PSO) algorithm is a heuristic optimization algorithm that is developed by Kennedy and Eberhart to simulate the behaviors of swarms, e.g. the birds flocking and fish schooling [12]. In recent years, PSO has been applied to deal with a large number of problems in robotics, such as model parameter identification, controller optimization design, trajectory generation and path planning, etc.

In PSO, several particles are defined to express the candidate solutions of the problem. The particles of the swarm fly around to search the optimal solution in the search space. For each particle, the velocity and position are updated iteratively in each iteration as follows:

$$\begin{aligned} v_i^{k+1} &= wv_i^k + c_1 r_1 (x_{pi}^k - x_i^k) + c_2 r_2 (x_g^k - x_i^k), \quad (5) \\ x_i^{k+1} &= x_i^k + v_i^{k+1}, \quad (6) \end{aligned}$$

where v_i^k and x_i^k stand for the velocity and position of the i th particle at the k th iteration; x_{pi}^k and x_g^k are the best position of the i th particle ($pbest$) and best position of the global particle swarm ($gbest$) until the k th iteration; w denotes the inertia weight; c_1 and c_2 are respectively indicate the acceleration

coefficients or cognitive and social parameters; r_1 and r_2 are two randomly generated numbers uniformly distributed on $[0, 1]$.

Up to now, a great number of modified PSO algorithms have been proposed to improve the performances of the above classical PSO algorithm, e.g. PSO with linearly decreased inertia weight (PSO-LDIW) [13], PSO with time-vary acceleration coefficients (PSO-TVAC) [14], PSO with constriction factor (PSO-CK) [15], adaptive PSO (APSO) [16], switching PSO (SPSO) [17], switching delayed PSO (SDPSO) [18], etc. In this paper, the MDPPO (multimodal delayed PSO) developed in [10] is applied to plan global smooth path for mobile robots. The equations to update the velocity and position of the MDPPO are presented as follows:

$$\begin{aligned} v_i^{k+1} &= wv_i^k + c_1 r_1 (x_{pi}^k - x_i^k) + c_2 r_2 (x_g^k - x_i^k) + \\ &\quad s_i^k c_3 r_3 (x_i^{k-\tau_i^k} - x_i^k) + s_g^k c_4 r_4 (x_g^{k-\tau_g^k} - x_i^k), \quad (7) \\ x_i^{k+1} &= x_i^k + v_i^{k+1}, \quad (8) \end{aligned}$$

where the inertia weight w is updated as equation (9):

$$w = (w_1 - w_2) \times \frac{k_{max} - k}{k_{max}} + w_2, \quad (9)$$

the acceleration coefficients c_1 and c_2 are updated as equation (10) and (11):

$$c_1 = (c_{1i} - c_{1f}) \times \frac{k_{max} - k}{k_{max}} + c_{1f}, \quad (10)$$

$$c_2 = (c_{2i} - c_{2f}) \times \frac{k_{max} - k}{k_{max}} + c_{2f}, \quad (11)$$

r_i ($i = 1, 2, 3, 4$) are the randomly produced numbers uniformly distributed in $[0, 1]$; s_i^k and s_g^k are the factors of intensity for the newly added terms in velocity updating equation; τ_i^k and τ_g^k are the random integer delays uniformly distributed in $[0, k]$ for the delayed $pbest$ and $gbest$; and c_3 and c_4 are set equal to c_1 and c_2 respectively without loss of generality.

Specially, the parameters of the newly added terms are determined according to TABLE I, where rand is the random number generating function which is used to produce the random number uniformly distributed in $[0, 1]$; the symbol $\lfloor \cdot \rfloor$ denotes the rounding down function; the intensity factors s_i^k and s_g^k are determined by the evolutionary state ξ^k and evolutionary factor E_f^k in each iteration. The evolutionary factor of the k th iteration denoted as E_f^k is computed by

$$E_f^k = \frac{d_g^k - d_{min}^k}{d_{max}^k - d_{min}^k}, \quad (12)$$

where d_{max}^k and d_{min}^k are the maximum and minimum mean distance between one particle to others in the swarm respectively; and d_g^k indicates the mean distance of the $gbest$. The mean distance of the i th particle of the k th iteration denoted as d_i^k can be computed by

$$d_i^k = \frac{1}{S-1} \sum_{j=1, j \neq i}^S \sqrt{\sum_{k=1}^D (x_i^k - x_j^k)^2}, \quad (13)$$

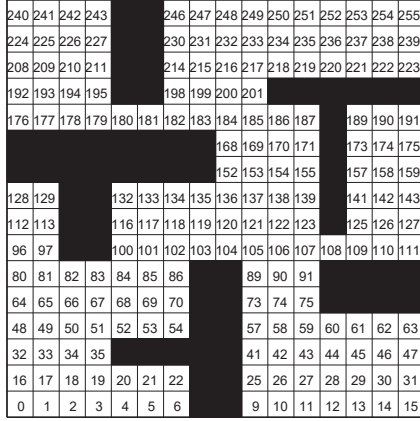


Fig. 1. Workspace of mobile robots.

where S and D stand for the size of swarm and the dimension of one particle respectively. In this paper, the evolutionary state is classified by using the following equal division strategy:

$$\xi^k = \begin{cases} 1, & 0.00 < E_f^k \leq 0.25, \\ 2, & 0.25 < E_f^k \leq 0.50, \\ 3, & 0.50 < E_f^k \leq 0.75, \\ 4, & 0.75 < E_f^k \leq 1.00. \end{cases} \quad (14)$$

TABLE I
PARAMETERS FOR DELAYED INFORMATION OF MDPSO

Mode	s_i^k	s_g^k	τ_i^k	τ_g^k
$\xi^k = 1$	0	0	—	—
$\xi^k = 2$	E_f^k	0	$[k \cdot \text{rand}]$	—
$\xi^k = 3$	0	E_f^k	—	$[k \cdot \text{rand}]$
$\xi^k = 4$	E_f^k	E_f^k	$[k \cdot \text{rand}]$	$[k \cdot \text{rand}]$

III. SMOOTH PATH PLANNING OF MOBILE ROBOTS

A. Environment Modeling

The workspace of the mobile robots is a two dimensional plane in this paper. The aim of the global smooth path planning is to find an optimal feasible smooth path for the mobile robots from the start to the target positions in the workspace shown in Fig. 1. In this paper, the workspace is divided into 16×16 grids, and each grid is a 10×10 units square, where the white and black grids indicate the feasible and infeasible space respectively. The obstacles in the workspace are extended with the minimum safe range according to the size of a mobile robot, and thus a mobile robot within the workspace can be regarded as a simple unit point.

B. Problem Statement

Consider a wheeled mobile robot (WMR) with the following nonholonomic motion model:

$$\begin{cases} \dot{x}(t) = v(t) \cdot \cos \theta(t) \\ \dot{y}(t) = v(t) \cdot \sin \theta(t) \\ \dot{\theta}(t) = \omega(t) \end{cases} \quad (15)$$

where x and y stand for the position coordinates; θ denotes the heading angle; v and ω indicate the linear and angular velocities of the mobile robot respectively.

In this paper, the MDPSO is combined with a continuous Bezier curve (instead of several connected Bezier curve segments) to plan a feasible, smooth and shortest path for mobile robots. Generally, a Bezier curve path is determined by a sequence of control points and the smoothness is usually featured by the curvature along the path, hence the planning of global smooth path is to look for a sequence of control points which define a feasible and shortest Bezier curve path with minimum curvature along the path. The objective function of the optimization are expressed as follows:

$$\begin{aligned} \min \quad & J = w_1 \times \|\mathbf{p}(t)\| + w_2 \times N_o \\ & + w_3 \times \max |\kappa(t)|, \quad t \in [0, 1], \\ \text{s.t.} \quad & \mathbf{p}(t) \in W_f, \\ & \mathbf{p}(t) \in C^2, \end{aligned}$$

where $\|\mathbf{p}(t)\|$ represents the path length of the continuous Bezier curve; N_o denotes the amount of infeasible path points; $\max |\kappa(t)|$ indicates the maximum absolute of the curvature on the path; W_f denotes a set of feasible spaces composing of the white square grids; C^2 denotes a set of second-order differentiable functions; and w_i ($i = 1, 2, 3$) are, respectively, the weights of each part of the objective function.

C. Simulation Results and Evaluation

In this section, the strategy of combining MDPSO and continuous Bezier curve has been applied to the global smooth path planning of mobile robots. The configuration of the simulation experiments are set as follows: the size of the swarm S is set as 100; the maximum iteration k_{max} is set as 100; the dimension of the particle D (which denotes the amount of control points of the continuous Bezier curve) is set as 7; the weights of the objective function w_i ($i = 1, 2, 3$) are set as 1, 50 and 100, respectively.

Several simulation experiments have been implemented, and two results of the experiments are demonstrated in Fig. 2 and Fig. 3 to illustrate the success of the proposed approach. In above two experiments, the start and target positions are set as (5, 5) to (155, 155) and (155, 5) to (5, 155), respectively. The blue circles indicate the position of the control points, and blue solid lines consist of the convex hull of the Bezier curve. The red solid curves indicate the optimum smooth path with continuous curvature on the whole path. For comparison, the counterpart results produced by using MDPSO and piecewise Bezier curve [10] are also depicted in Fig. 4 and Fig. 5. The curvature of the paths is described for the two cases in Fig. 6 and Fig. 7, respectively. It is obviously that the curvature variations of the paths produced by using MDPSO and continuous Bezier curve are continuous rather than their counterparts which have several steps on the curvature variations. Besides, the curvature variations of the smooth paths in this paper are also milder than the ones in [10]. Because the curvature variations are continuous along the planned smooth path, the frequent mode switches of the movement (e.g. stop, rotation

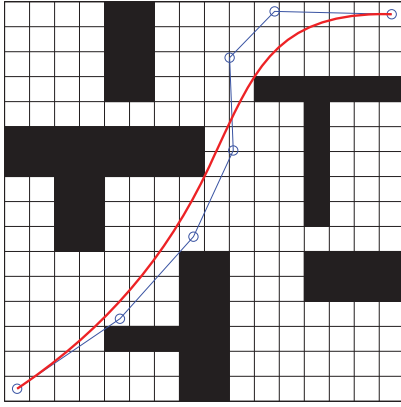


Fig. 2. Smooth path I (Continuous Bezier curve).

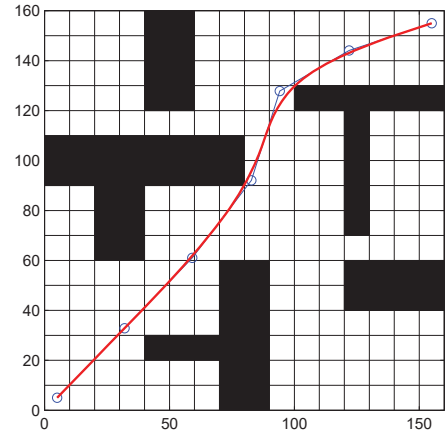


Fig. 4. Smooth path I (Piecewise Bezier curve).

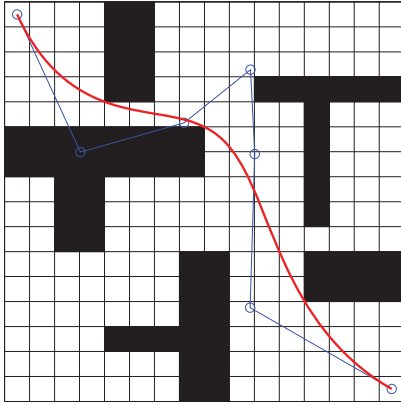


Fig. 3. Smooth path II (Continuous Bezier curve).

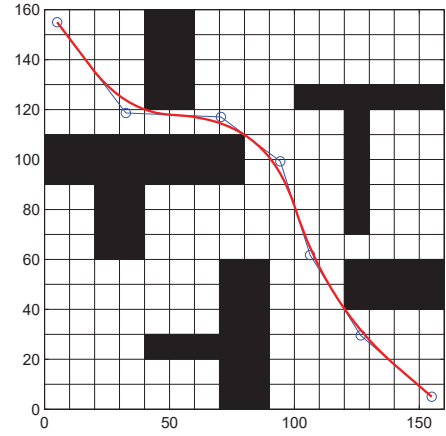


Fig. 5. Smooth path II (Piecewise Bezier curve).

and restart) are avoided and thus the smooth movements of mobile robots are ensured.

Note that the issue of this paper is challenging for at least the following reasons. Firstly, a feasible optimum path is located in the narrow valley composed of nearby obstacles. Secondly, the continuous Bezier curve is included in a convex hull, where the control points are not interpolated by the Bezier curve. This property makes the strategy of path smoothing after a linear path planning not workable, and that is why the piecewise curves are usually used in smooth path planning problem. Thirdly, minimum curvature is linked to the performance indices, which makes the smooth path planning a multimodal optimization problem that frequently trapping into local optima. Fortunately, the MDPSO could successfully complete the challenging task, which have been verified by the simulation results.

IV. CONCLUSIONS

In this paper, a new strategy combining MDPSO and continuous Bezier curve has been developed to plan global smooth path for mobile robots. With this strategy, the curvature of the path is ensured to be continuous and the MDPSO is verified to be able to deal with the obstacles of the optimization issues successfully.

In the future work, we will focus on many interesting and complicated topics [19], [20], e.g. multi-robot path planning, dynamic path planning, path planning with constrained kinematic dynamics and 3-D path planning, etc.

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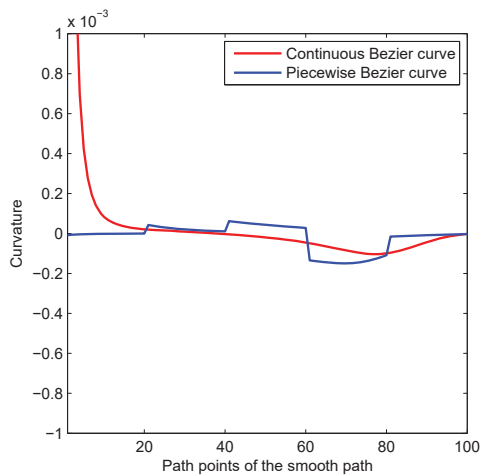


Fig. 6. Curvature comparison of smooth path I.

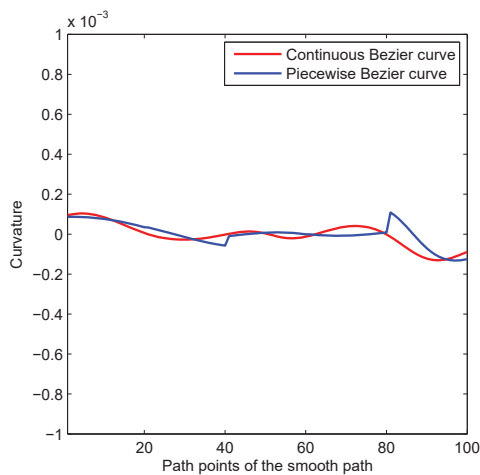


Fig. 7. Curvature comparison of smooth path II.

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